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Hello, I'm Priyanshu Kumar, currently pursuing BS in Data Science and Applications from IIT Madras. I'm currently in Diploma Level and this is my Modern Application Development 1 Project – "Vehicle Parking App - V1".

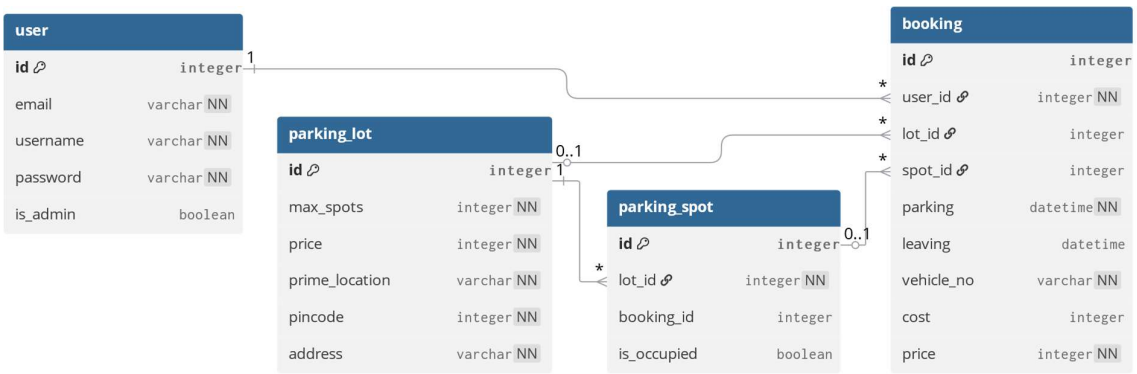
Description

In this project we need to create a Multi-user (User and Admin) app, where Admin manages the Parking Lots and Parking Spots for 4 – wheeler parking and User books a Parking Spot in a Parking Lot or Releases the currently booked Parking Spot.

Technologies used

Flask – for application back-end
Jinja2 – for HTML templating
Bootstrap – for front-end styling
SQLalchemy – for database ORM
Flask-SQLalchemy – Flask extension for simplifying SQLalchemy Operations
SQLite – database for storing and managing data

DB Schema Design



user Table - To store different types of users (Admin, User), *is_admin* attribute to differentiate between them.

parking_lot Table – To store different Parking Lots by ID and their information.

parking_spot Table – To store all the Parking Spots of Parking Lots. *lot_id* attribute to differentiate to which Parking Lot, a Parking Spot belongs to, *is_occupied* attribute to know if a Parking Spot is occupied or not and *booking_id*

attribute to store the Booking ID of current booking or last booking.

booking Table – To store all the different bookings by ID with User ID, Lot ID and Spot ID. *parking* attribute to store Parking time, *leaving* attribute to store Leaving time, *vehicle_no* attribute to store Vehicle No. of the parked vehicle, *cost* attribute to store the total Cost after parking out and *price* attribute to store the /hr price of the Parking Lot at the Time of Booking.

Architecture and Features

Project Folder Structure ↴

```
vehicle-parking-app-v1\
│
├─── application\
│    ├─── controllers.py
│    └─── models.py
│
├─── instance\
│    └─── parking.db
│
├─── static\
│    └─── {{ IMGs, ...etc. }}
│
├─── templates\
│    └─── {{ all the HTML Templates }}
│
├─── app.py
└─── {{ Other Files }}
```

Controllers and models are in the *application* folder, and all the HTML templates are in the *templates* folder. The database *parking.db* is programmatically created in the *instance* folder. The *app.py* file is in the root directory which creates the database, Flask app instance and runs the app.

Features:

Login – Single login form is used to login both user and admin. user and admin are identified by *is_admin* attribute in the database.

Admin – Admin can Edit or delete a Parking Lot or spot (if not occupied), Admin can add any number of parking lots and spots, Admin can view all the registered users and all the bookings and summary of revenue from each parking lot and Admin can search users by email or name and parking lots by location or pincode.

User – A user can view all the parking lot or search a parking lot by location or pincode and book a spot in the parking lot (if available), user can release an occupied spot while the total cost is calculated by the difference between parking time and leaving time, and user can view the history of every parking and summary of how many times the user had booked a spot in a parking lot.

Video

Video >>> here